KATHMANDU UNIVERSITY

SCHOOL OF ENGINEERING

DEPARTMENT OF MECHANICAL ENGINEERING

A CASE STUDY REPORT ON

TECHNICAL FEASIBILITY STUDY OF SOLAR WATER PUMPING SYSTEM: A CASE STUDY OF SUNKOSHI 1123M PROJECT

In Partial Fulfillment of the Requirements for the

Course of Renewable Energy Core I (MEEG 428)

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1. Introduction

Water is essential for every living being on earth. It supports every function of life including agriculture, industrial purpose, ecosystems and human health. However, increasing demands from population growth, climate change, and pollution place negative impact on water resources. This will produce scarcity of drinking water on future.

There are several sources of water available on earth. Surface water, ground water and atmospheric water are the major sources of it. For human being, water is needed in different ways as to meet this need, they use different types of techniques to extract water from different sources. Ground water extraction, Rain water harvesting, surface water extraction, solar water pumping, atmospheric water generation are some daily practicing methods of water extraction.

In places like upland community with lower elevation water sources of Nepal, which is located high up in the hill, getting water can be tough. There are some small natural water sources like springs, tiny streams, and underground water stores but due to sudden change in the climate, global warming, deforestation and pollution clean drinking water is completely disappeared from these places. So, to make water available around this place some kind of change technique should be applied. For this place solar water pumping will be the best to supplied water from lower elevation to higher one.

One of the best self-sustaining renewable sources to extract water in place like upland community with lower elevation water sources is the solar water pumping system. It uses solar thermal energy to power up the water pump, making easy for remote and off grid areas of Nepal. In place for high head water pumping system, it is more advantageous to use the solar power for water pumping. Due to its high head more power of solar system should be design, but in offload condition of water pumping time grid uses solar thermal power from the system which will also cost benefit.

2. Background And Literature Review

As Nepal is best known for rich in water resources as compare to area around the world. In Nepal approximately around 6000 rivers and rivulets are flows within drainage area of 194,471km^2. [1] Though Nepal has rich in its water resources, high topography region stills lacks clean sources of water. Some places of Nepal like Bhumisthan-3, Argakhanchi District still they have to wait 30-40 minutes to fill a 20L bucket or jar. Dallu-4, Dailekh District also have the same kinds of problem of water. [2]

Besides that, scarcity of drinking water is also found at a village called Bhangeri which is located at Ramechhap district of Nepal. It lies in eastern part of Nepal and central part of Ramechhap district. Its area is about 1.61km^2 and altitude of about 1400m form sea level. [3]

Figure 1 Map of Bhangeri village of Ramechhap District [3]

Problem of Unavailability of clean water in rural Ramechhap.

Harsh climate, Deforestation, Pollution, 2072 BS Earthquake and geographical condition are the major factor for unavailability of the clean water in this village.

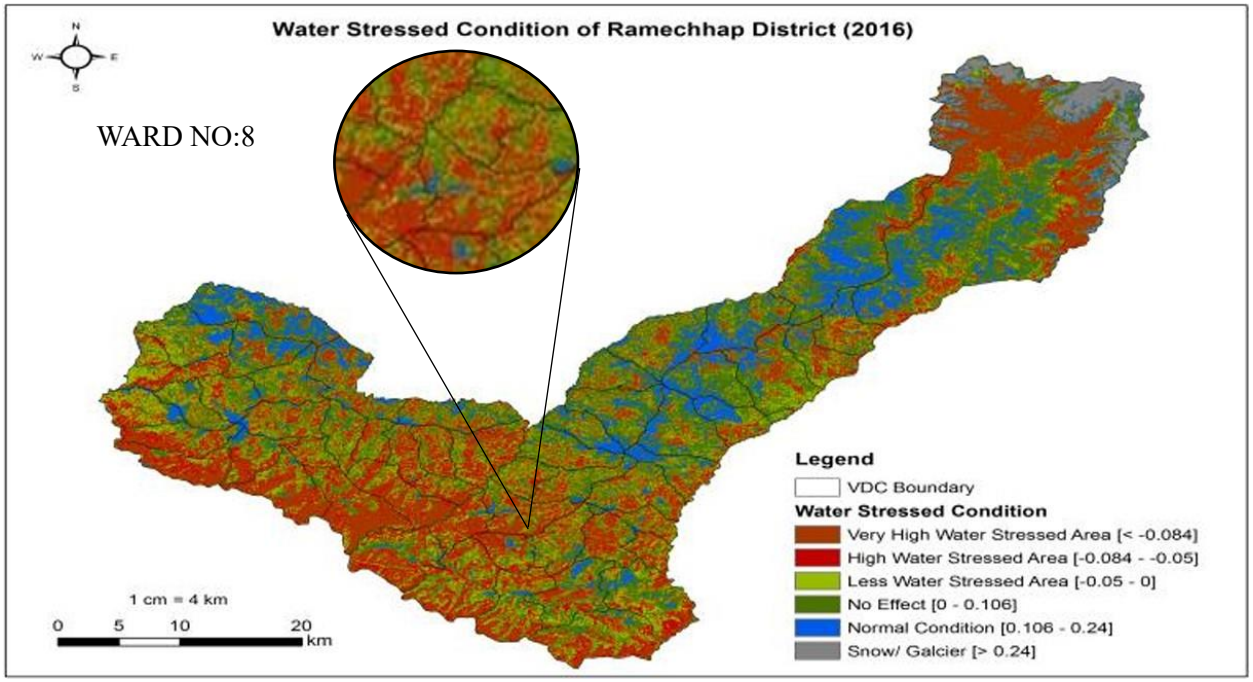

Figure 2 Water stress condition of Ramechhap District [4]

The above figure shows the water stress condition of the Ramechhap district. On the highlighted portion its show the very high water stressed area which is Ramechhap municipality. Due to sudden drought, most of the water sources have been dried which led to increased number of seasonal migrations. According to the Office of District Coordination Committee, GoN, Ramechhap recorded total of 23 permanent migrationin 2015/2016. In the past 5-7 years period the major sources of water in Pakarbas and Chisapani have already dried. Also Gadapani, kakale padhero in sukulman village, Raamchemuhan in Ramche village, Sanne padhera, Munitar in Sunnegau, archalle in Gaude and in Chisapani VDC thulopadhero, Aitabare, Ganesthan padhera etc.which were the major sources had dried in recent years. [4] The main problems for the disappearing of water sources are deforestation, increase in temperature, sudden change in climatic condition, recent earthquake 2072 BS and high-altitude community settlement.

3. Methodology

Technical Analysis

Site information

Bhangeri village (27°19'57"N 86°04'29"E) is situated at Ramechhap Municipality Ward No- 8, Ramechhap District, Nepal. The village is surrounded by various geographical landmarks. To the east of the village lies Ramechhap Bazar, while to the west, there is Dukursing Danda. Babiyakharka is located to the north of the village, and Lyanglyang is situated to the south. The geographical land of Bhangeri village is stepped down, inclining towards the east side of the village, where elevation ranges from 454m to 1577m from sea level (google earth). Water is pump from elevation of 407m which is in range of Sunkoshi River this place is called Birta Besi (27°17'37"N 86°03'24"E) to Dukursing danda (27°20'00"N 86°03'41"E) an elevation of 1577m. [3]

Identification of water requirement

To design solar water pumping for a particular site it is necessary to identify the water demand of that place. Water demand can be defined as the aggregation of water usages in different sector of that particular places and express as in terms of liters/day. By estimating demand, the water that pump from the source is distributed to 5 communities Bhangeri, Chhap Simanti, Lyanglyang, Upper sites of Babiyakharka and some portion of Manedada. Among these places some important place including Schools, Hotels, and Residential houses are taken.

All the data recorded below in table are of only 5 communities of 5 houses from Ramechhap Municipality Ward No 8.

Table 1 Quantity of water requirement per day of 5 houses of Bhangeri Village

S.No.	Activities	Quantity of Water in House No:1 Liter/Day	Quantity of Water in House No:2 Liter/Day	Quantity of Water in House No:3 Liter/Day	Quantity of Water in House No:4 Liter/Day	Quantity of Water in House No:5 Liter/Day	References
1	Drinking	10	20	25	5	20	
2	Cooking	20	15	30	10	30	
3	Washing	100	100	200	50	100	
4	Toilets	50	50	100	20	100	
5	Bathing	50	200	200	10	100	
6	Cattle	50	100	300	50	500	
7	Cleaning	50	75	100	20	150	
S.No.	**Activities**	**Quantity of water used liters/day**					
1	Hotels	1000					
2	School	5000					

According to census of 2011 AD the total population of Ramechhap municipality was 4,869. [5] and number of households are 830 from this household's data we take average of 5 data and calculated the 830 households including separate data of hotels and schools. [6]

Table 2 Water requirement per day of 830 households.

Activities	Average Quantity of water required
Basic Hygiene (Hand washing, Toilet, Bathing)	(95.33*830) =79126
Drinking	(16*830) =13280
Cooking	(21*830) =1743
Cleaning	(79*830) =65570
Schools (2)	(5000) *2=10000
Hotels (15)	(1000*15) =15000
Total Water Required per day	184719 liters per day

Water resources

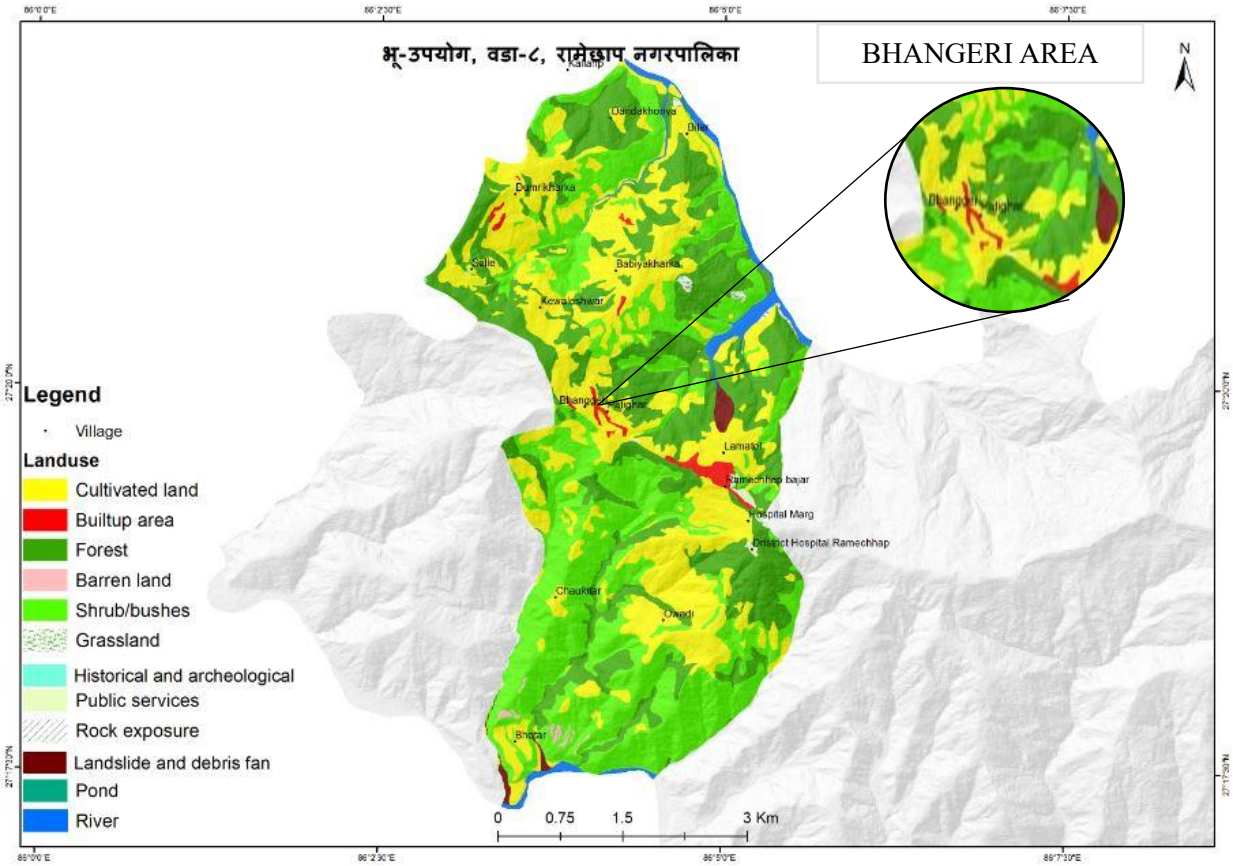

Figure 3 Land Use and Land Cover (LULC) map of Ramechhap municipality [5]

Above figure describes the land use and land cover map of Ramechhap Municipality Ward No: 8. It is clearly observed there is no any sign of river and pond found in this area, only in the lower part of map river can be observe which is a Sunkoshi River.

Before 2058 BS, there were different kinds of water sources available Kartike Stream, Dharapani forest Stream, Gadapani forest stream, Pangrepani forest well and Chhatiwoni River and more water resources were found inside dense forest. Later, 2058 BS Chhatiwani River water was distributed through water pumping system. As now all this above mention water sources were dried due to drastic change in climatic

condition, deforestation and the recent 2072 earthquakes was also main factor for disappearing of water resources except Chhatiwani River.

Water storage

Table 3 Quantitative analysis of water storage

S.No.	Purpose of Tank	Volume of Tank (Liters)	Elevation (m)	Time to fill up (Hours)
1	Reserve	250000	454	8.56
2	Reserve	200000	662	6.84
3	Reserve	200000	877	6.84
4	Reserve	200000	1097	6.84
5	Reserve	250000	1317	8.56
6	Supply	250000	1577	8.56

Selection of Water Pump

The water pump should coverage the head of each tank and also should have the good amount of discharge rate. The specification of the motor used is listed below:

Table 4 Specification of Water pump

Power	Input Volt.	Flow	Head	Inlet	Outlet
20hp, 18.5Kw	3 phase, AC 220V-480V	$29.2m^3/hr$	103m	50mm	50mm

Operation water pumping method

As there are 2 water pumps install in parallel combination for each water tank. This will increase the head of pump to (2*103) m but the discharge rate remains constant as $29.2m^3/hr$.

Calculation

Calculation for time to fill up tanks

Volume of Tank= 250000 liters

Volume of tank= 200000 liters

Discharge of Water pump= $29.2m^3/hr$

Time requires to fill up tank= $\frac{250000}{29200}$

= 8.56 hours

Time requires to fill 200000 liters tank= $\frac{200000}{29.2}$

= 6.84 hours

The daily water requirement of communities= 184719liters

As per the daily requirement of communities the water pump should run= $\frac{184719}{29.2}$

= 6.3 hours

Calculation for water supply to community.

Average head of Area where water to be supply= 50m

Diameter of Pipe= $50*10^{-3}$m

Area of Pipe= $1.96*10^{-3}m^2$

Discharge for Bhangeri village= $A*\sqrt{2gh}$

$=1.96*10^{-3}*\sqrt{2}*9.8*50$

$=0.06135m^3/s$

=226800ltr per hour

Note: there will be the difference in height for different communities and houses above calculation is based without losses.

Solar resources feasibility

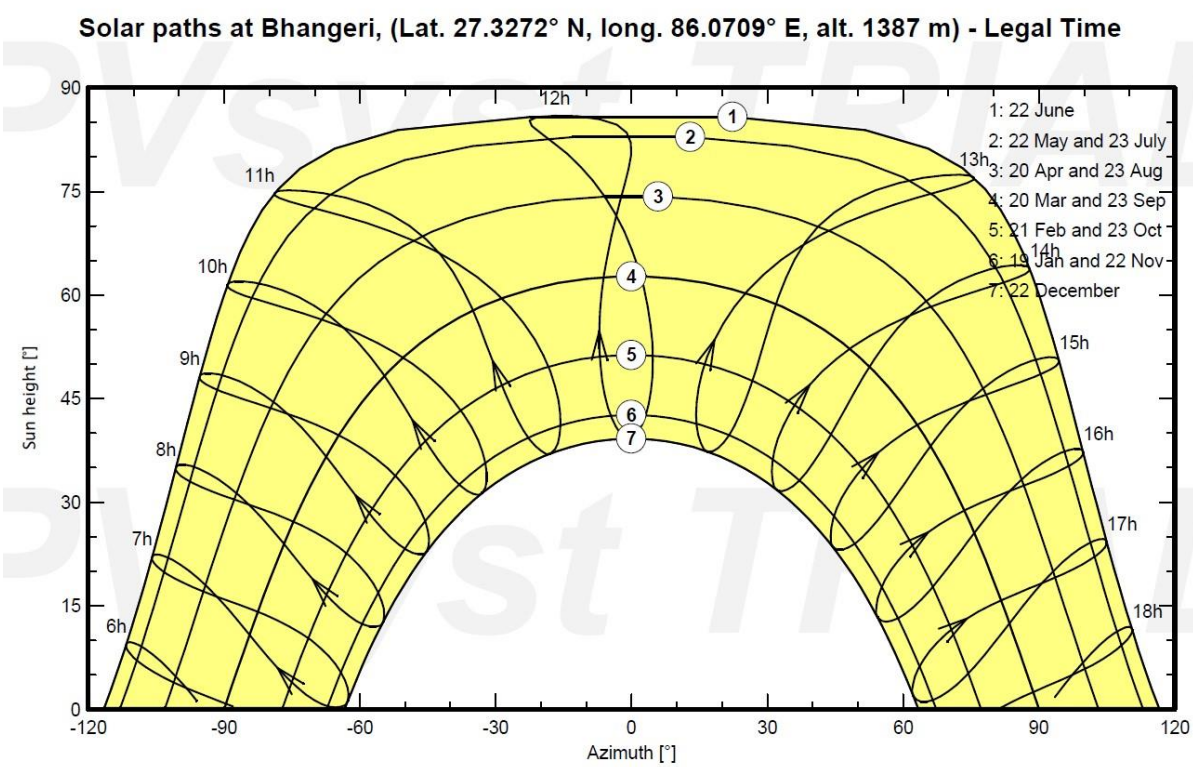

Figure 4 Sun path diagram of Bhangeri Village [7]

This is the sun path diagram of Bhangeri village. This shows that the sun will be at highest peak of the sky at 22 June and lowest at 22 December. On month of June the radiation of sun that start to operate the solar panel starts at about 7 AM and set about at 6 PM.

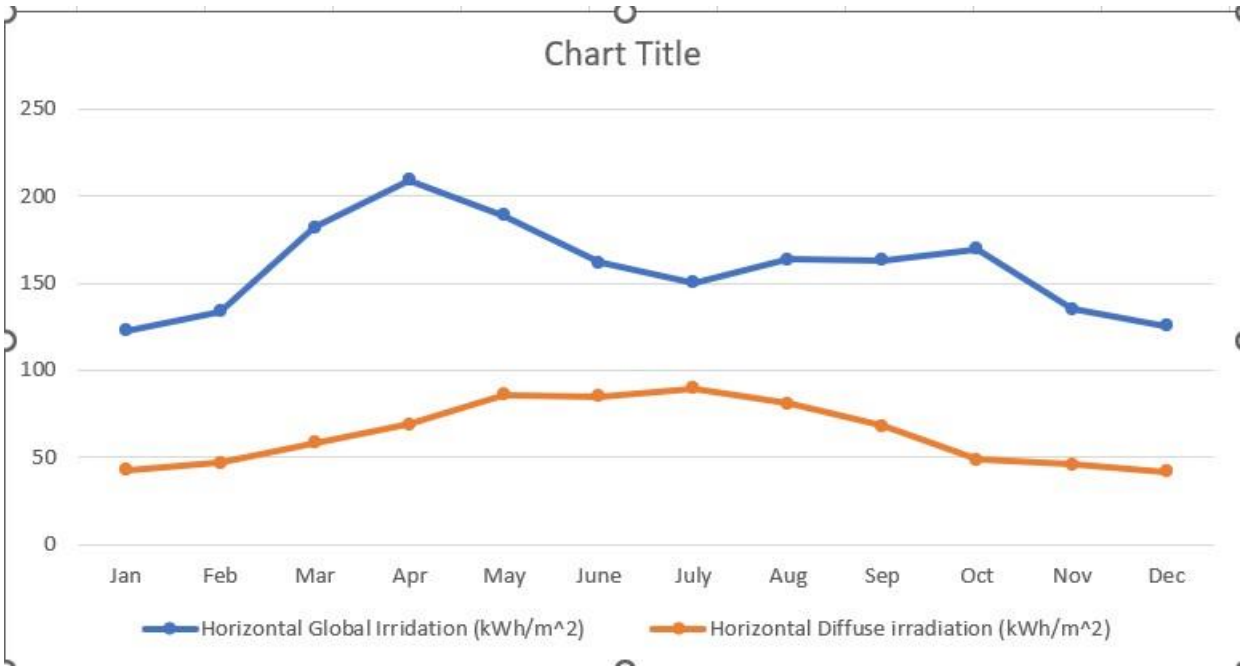

Figure 5 Graph showing Irridation of Bhangeri Village

This is the chart showing Global Irradiation of Bhangeri Village. The chart shows that on month of Mar to June the Horizontal global irradiation is high and low on month of January and dec. the Horizontal diffuse irradiation was high on month of May to September and low on January and December.

PV Array Characteristics

PV module		Inverter	
Manufacturer	Generic	Manufacturer	Generic
Model	TSM-DE09R-08W-420wp	Model	Solarmax 100TS
(Original PVsyst database)		(Original PVsyst database)	
Unit Nom. Power	420 Wp	Unit Nom. Power	100 kWac
Number of PV modules	481 units	Number of inverters	2 units
Nominal (STC)	202 kWp	Total power	200 kWac
Modules	37 Strings x 13 In series	Operating voltage	430-800 V
At operating cond. (50°C)		Pnom ratio (DC:AC)	1.01
Pmpp	185 kWp		
U mpp	495 V		
I mpp	374 A		
Total PV power		**Total inverter power**	
Nominal (STC)	202 kWp	Total power	200 kWac
Total	481 modules	Number of inverters	2 units
Module area	961 m²	Pnom ratio	1.01

PVsyst TRIAL

Figure 6 Solar PV array Characteristics [7]

The above figure shows the PV array model for operation of water pumping system. The above system is designed using the PVsyst software. The maximum power output is calculated by calculating the number of water pump used and by using its operated power ratings. This design is based on full on operating of 10 water pump.

Optimization of Design.

As per the regular supply of water pumping system min of 4 water pump can be used daily. And also, we can operate only 4 water pumps to minimize the power uses. On rest of the period, we can also connect this system to the grid.

4. Result

Technical analysis

In order to full fill the water demand of the Ramechhap Municipality Ward No. 8 of certain communities including Bhangeri Village, some part of Manedada, upper portion of Babiyakharka, and Chhap Simanti solar water pumping system will help. This water pumping system has total of 6 water tank 5 are reserve tank and 1 is supply tank. The water pump will operate as in step order to fill the tank. In order to power up the whole systems each water pump rated 18.5kw total of 10 water pump are install on its full potential it needs power of 185kw. But the full system will not run once all the tanks are full. Only minimum of 4 water pumps will run stepwise to fill the tanks.

The daily water requirement of the above mention places is about 184719 liters per day. The water will distribute in communities in order way as one day for Bhangeri and Manedada and another day Chhapsimanti and Babiyakharka. As we divided the water demand Each day distribution demand will be 92359.5 liters per day. 226800 liters of water is distributed each day within 1 hour of morning time which is enough for the uses of 2 days for 2 communities. Due to this process the time the water pump running will be reduce as per the volume of water needed.

Table 5 Design Parameter of Solar PV for water pumping

S.No.	Module Name	Number of units	Power Output (kWp)	Voltage (Volt)	Current (amp)	Area Required (m^2)
1	PV module	481	185	495	374	961
2	Inverter	2	2*100	430-800	-	-
3	Water Pump	10	185	220-480	-	-

As for the solar pumping process this village receives around 158.84 kWh/m^2 monthly radiation. Based on this data on installing each module of 420Wp we require 481 units of solar pv with modules at 37 strings*13 in series that produce power of 185kWp which will require 961m^2 Area to install it, Inverter will convert this power in to 200kWp of AC current.

5. Discussion

Once all the tanks are full, we shouldn't have to run all the water pump so, to optimize the power only the power needed for 4 water pumps i.e. 74kWp is reserve on running the system. We can connect the rest of the power to the grid.

6. Conclusion

The techno-economic feasibility study of the solar water pumping system for the Sunkoshi 1072m project in Bhangeri village, Ramechhap District, highlights the technical viability and environmental benefits of using solar energy to address water scarcity in remote, upland communities. The study confirms that the system, with its design of 10 water pumps powered by 481 PV modules, can meet the daily water requirement of 184,719 liters for multiple communities by efficiently utilizing the substantial solar radiation available in the region. This place is technically feasible and capable of addressing the community's water scarcity issues effectively. The system, comprising 481 PV modules arranged in 37 strings of 13 modules each, generates a total power output of 185 kWp, requiring an installation area of 961 square meters. This design leverages the substantial solar radiation of approximately 158.84 kWh/m^2 received monthly by the village, ensuring efficient energy capture and conversion. Two inverters, each with a capacity of 100 kW, convert the generated power to 200 kWp of AC current, which is used to operate 10 water pumps. These pumps are strategically managed to optimize energy use while meeting the daily water demand of 184,719 liters for the community. The successful implementation of this system not only provides a reliable water supply but also exemplifies the potential of renewable energy solutions in enhancing the quality of life in remote, water-stressed regions.

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